

Scalability & Future Plan

OptiCrop: Smart Agricultural Production Optimization Engine

Date	28 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how the current OptiCrop solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static dataset — model trained on fixed crop dataset with no real-time data ingestion	Recommendations may not reflect current soil or climate conditions; reduces accuracy for dynamic environments	High
2	Single-user Flask server with no concurrency support	Cannot handle multiple simultaneous users; crashes under load — limits production deployment	High
3	No persistent data storage — user	Cannot track historical	Medium

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
	inputs and results are not saved between sessions	recommendations or build user-specific profiles for personalized advice	

Scalability Plan

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-threaded Flask dev server; supports ~1 concurrent user	Deploy with Gunicorn + Nginx on a cloud VM (AWS EC2 / GCP); use load balancing to support 500+ concurrent users
2	Data Storage	No database; data lives in CSV files loaded at runtime	Integrate PostgreSQL or MongoDB to store user inputs, recommendations, and historical data; enable persistent session management
3	Performance	Model loaded fresh per request; no caching; response ~2–3 seconds	Cache trained model in memory using joblib; implement Redis for session caching; target response time under 500ms
4	Security	No authentication; all endpoints publicly accessible; no input validation	Add JWT-based user authentication, input sanitization, HTTPS via SSL certificate, and role-based access for farmers vs. researchers

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	IoT Sensor Integration — connect real-time soil sensors (pH, moisture, NPK) to auto-populate input fields and eliminate manual data entry	Q3 2026	Enables fully automated crop recommendations based on live field data; significantly improves accuracy and farmer convenience
Phase 2	Mobile Application (Android/iOS) — develop a farmer-friendly mobile app using React Native with offline support and regional language options	Q4 2026	Expands reach to rural farmers with limited desktop access; multilingual support increases adoption in diverse agricultural regions
Phase 3	Deep Learning Model Upgrade — replace Random Forest with an LSTM/CNN model trained on time-series climate data for seasonal predictions	Q1 2027	Higher prediction accuracy (target 99.5%+); supports multi-season planning and climate-resilient farming recommendations
Phase 4	Government & Research API — expose OptiCrop predictions via a REST API for integration with agricultural policy platforms and research databases	Q2 2027	Enables large-scale data-driven agricultural policy planning; positions OptiCrop as a national-level precision farming tool